**7. Scenario: You are working as a data analyst for an e-commerce company. You have been given a dataset containing information about customer orders, stored in a Pandas DataFrame named order_data. The DataFrame has columns for customer ID, order date, product name, and order quantity. Your task is to analyze the data and answer specific questions about the orders.**

**Question: Using Pandas DataFrame operations, how would you find the following information**

**from the order_data DataFrame:**

**1. The total number of orders made by each customer.**

**2. The average order quantity for each product.**

**3. The earliest and latest order dates in the dataset.**

```
import pandas as pd


order_data = pd.DataFrame({
   'Customer_ID': [101, 102, 101, 103, 102],
   'Order_Date': ['2026-01-05', '2026-01-10',
            '2026-02-15', '2026-02-20', '2026-03-01'],
   'Product_Name': ['Laptop', 'Phone', 'Phone',
             'Laptop', 'Tablet'],
   'Order_Quantity': [2, 1, 3, 1, 2]
})


order_data['Order_Date'] = pd.to_datetime(order_data['Order_Date'])


# Total orders by each customer
print("Orders per customer:")
print(order_data.groupby('Customer_ID').size())
```

```
# Average quantity per product
print("\nAverage order quantity per product:")
print(order_data.groupby('Product_Name')['Order_Quantity'].mean())


# Earliest and latest order dates
print("\nEarliest order date:", order_data['Order_Date'].min().date())
print("Latest order date:", order_data['Order_Date'].max().date())
```

```
=============== RESTART: C:/Users/Jannathul firdaus/exp 7 fods.py ==============
Orders per customer:
Customer_ID
101    2
102    2
103    1
dtype: int64

Average order quantity per product:
Product_Name
Laptop    1.5
Phone     2.0
Tablet    2.0
Name: Order_Quantity, dtype: float64

Earliest order date: 2026-01-05
Latest order date: 2026-03-01
```

**8. Scenario: You are a data scientist working for a company that sells products online. You have been tasked with analyzing the sales data for the past month. The data is stored in a Pandas data frame.**

**Question: How would you find the top 5 products that have been sold the most in the past month?**

```
import pandas as pd


sales_data = pd.DataFrame({
   'Product': ['Laptop', 'Phone', 'Tablet', 'Phone',
           'Laptop', 'Headphones', 'Phone', 'Tablet'],
   'Quantity': [5, 20, 12, 15, 8, 25, 10, 9],
   'Order_Date': [
      '2026-09-20', '2026-09-22',
      '2026-09-25', '2026-10-01',
```

```
        '2026-10-03', '2026-10-05',
        '2026-10-08', '2026-10-10'
    ]
})

sales_data['Order_Date'] = pd.to_datetime(sales_data['Order_Date'])

# Select orders from the past month (example period)
start_date = '2026-09-13'
end_date = '2026-10-13'

recent_sales = sales_data[
    (sales_data['Order_Date'] >= start_date) &
    (sales_data['Order_Date'] <= end_date)
]

# Find top 5 products by quantity sold
top_products = (
    recent_sales.groupby('Product')['Quantity']
    .sum()
    .sort_values(ascending=False)
    .head(5)
)

print("Top 5 best-selling products:")
print(top_products)
```

```
========== RESTART: C:/Users/Jannathul firdaus/exp 8 fods.py ==========
Top 5 best-selling products:
Product
Phone         45
Headphones    25
Tablet        21
Laptop        13
Name: Quantity, dtype: int64
```

**9. Scenario: You work for a real estate agency and have been given a dataset containing**

**information about properties for sale. The dataset is stored in a Pandas DataFrame named property_data. The DataFrame has columns for property ID, location, number of bedrooms, area in square feet, and listing price. Your task is to analyze the data and answer specific questions about the properties.**

**Question: Using Pandas DataFrame operations, how would you find the following information**

**from the property_data DataFrame:**

**1. The average listing price of properties in each location.**

**2. The number of properties with more than four bedrooms.**

**3. The property with the largest area.**

```
import pandas as pd


property_data = pd.DataFrame({
   'Property_ID': [1, 2, 3, 4, 5],
   'Location': ['Chennai', 'Mumbai', 'Chennai',
            'Delhi', 'Mumbai'],
   'Bedrooms': [3, 5, 4, 6, 2],
   'Area': [1500, 2500, 1800, 3000, 1200],
   'Listing_Price': [5000000, 9000000, 6500000,
               12000000, 4000000]
})
```

```
# Average listing price by location
print("Average listing price by location:")
print(property_data.groupby('Location')['Listing_Price'].mean())

# Count properties with more than 4 bedrooms
count = (property_data['Bedrooms'] > 4).sum()
print("\nProperties with more than 4 bedrooms:", count)

# Property with the largest area
largest = property_data.loc[property_data['Area'].idxmax()]
print("\nProperty with the largest area:")
print(largest)
```

```
========== RESTART: C:/Users/Jannathul firdaus/exp 9 fods.py ==========
Average listing price by location:
Location
Chennai     5750000.0
Delhi      12000000.0
Mumbai      6500000.0
Name: Listing_Price, dtype: float64

Properties with more than 4 bedrooms: 2

Property with the largest area:
Property_ID             4
Location            Delhi
Bedrooms                6
Area                 3000
Listing_Price    12000000
Name: 3, dtype: object
>
```